

CLA Reports Style Guide

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Style Guide

--- Revision History ---

[1.4.0] - 2026-01-07

Changed

- Verified PlotlyStyleManager references match current API implementation.
- Confirmed visualization standards align with codebase implementation.

[1.3.0] - 2025-12-20

Changed

- Updated Section 2.1 to include a "Redundancy Exception" for single-mode datasets.

[1.2.0] - 2025-12-20

Changed

- Renamed application to "Contest Log Analytics".
- Implemented "Three-Line Title" standard (Report, Context, Scope).
- Added "Metadata Header/Footer" standards for Text and Visual reports.

[1.1.0] - 2025-12-06

Added

- Added "The Hierarchical Aggregation Standard" to prevent misleading sums.
- Added "Explicit Scope Labeling" to clarify data subsets in comparisons.
- Added "General UX Principles" (Smart Suppression).

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[1.0.0] - 2025-12-06

Changed

- Major update to support Plotly interactive charts and dashboard layouts.

1. Introduction

This document is the single source of truth for the visual and formatting standards of all report outputs generated by the **Contest Log Analytics (CLA)**. Its purpose is to ensure that all reports, regardless of type, have a consistent, clean, and professional appearance. All new reports must adhere to these guidelines.

1.1. General UX Principles

- **Signal-to-Noise Ratio:** Maximize signal, minimize noise. Reports should prioritize "Actionable Intelligence" over raw data dumps.
 - **Smart Suppression:** If a "Detail Block" provides no new information compared to its parent "Summary Block" (e.g., a Single-Mode band where Band Total == Mode Total), the Detail Block **should be suppressed** to reduce clutter.
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2. Report Headers & Titles

2.1. File Titles (Top of Document)

All generated reports must use the standard **Three-Line Title** format.

- **Line 1 (Report Name):** The official name (e.g., "Cumulative Difference Plots").
- **Line 2 (Context):** The event context (e.g., "2025 NAQP CW AUG - K1ABC vs W1XYZ").
- **Line 3 (Scope):** The data scope. **Mandatory.** (e.g., "All Bands", "20 Meters", "20 Meters [CW]").
 - *Exception:* The Mode label should be omitted if the dataset contains only a single mode (e.g., in a CW-only contest).

2.2. Branding & Metadata (Headers/Footers)

Every report must include the application signature and CTY provenance.

Text Reports (Upper Right Header) Aligned to the right margin of the content block:

Contest Log Analytics by KD4D
CTY File: YYYY-MM-DD CTY-XXXX

Visual Reports (Bottom Center Footer) Centered below the plot area:

Contest Log Analytics by KD4D
CTY File: YYYY-MM-DD CTY-XXXX

2.3. Section Headers (Internal Dividers)

For reports that contain multiple distinct blocks or tables, strict visual separation is required.

- **Separator:** A section must begin with a full-width visual separator (e.g., 80 dashes --- in text reports).
 - **Header Text:** The section title must be centered relative to the content block it precedes.
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3. Text Reports (.txt)

This section defines the standards for all plain-text reports.

3.1. Library Standards

The creation of text-based tables is governed by the following two-tool approach:

- **For Complex Reports (prettytable):** Recommended for fixed-width layouts, precise column alignment, or "stitching" multiple tables.
- **For Simple Reports (tabulate):** Suitable for single, standalone tables where complex alignment is not a concern.

3.2. The Drill-Down Pattern

To ensure scannability, text reports must follow the "Drill-Down" structure:

1. **Executive Summary:** The first block must be a high-level summary (e.g., All Bands, Total Score).
2. **Sequential Details:** Subsequent blocks should break the data down by specific segments (e.g., Band-by-Band Details), ordered logically (e.g., 160M to 10M).

3.3. Block Layout Standards

Unlike graphical plots, text reports must **NOT** be globally centered on the page/terminal.

- **Left Anchor:** All tables and content blocks must be anchored to the left margin (Column 0).
- **Relative Centering:** Table titles and column headers must be centered relative to the *width of the specific table*, not the global document width.
- *Correct:* Title is centered over the 60-character table below it.
- *Incorrect:* Title is centered at character 40 of an 80-character terminal while the table sits at the left margin.

3.4. The Hierarchical Aggregation Standard

When presenting aggregated data (e.g., Band Totals) alongside constituent parts (e.g., Mode Counts), the report **must** use a Parent/Child layout to prevent misleading mental arithmetic (e.g., "10 + 10 = 20" when the union is actually 12).

- **Structure:**
 - **Parent Row:** Labeled ALL or TOTAL. Displays the **Union** (deduplicated total).
- **Child Rows:** Indented. Display strict, specific counts for that subset (e.g., CW, PH).
- **Visual Cue:** Parent rows must be visually distinguished (e.g., via a separator line ----- or all-caps label) to signal they are summaries, not addends.

3.5. Explicit Scope Labeling

To prevent "Visual Lies" where a user mistakes a subset statistic for a global statistic, column headers must explicitly state the **Scope** of the data if it is a subset.

- **Mandate:** Do not use generic labels like "Run" or "Total" if they refer to a filtered subset (like "Unique Run").
 - **Pattern:** Use "Super-Headers" or compound labels.
 - *Bad:* | Run | S&P | (inside a Unique table)
 - *Good:* | UNIQUE (Run) | UNIQUE (S&P) | OR | Unique Run | Unique S&P |
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4. HTML Reports (.html)

This section defines the standards for all web-based reports.

4.1. Styling Framework

All styling for HTML reports must be implemented using **Tailwind CSS**. No custom CSS in <style> blocks should be used unless absolutely necessary.

4.2. Layout & Table Styling

HTML reports often act as dashboards. Therefore, strict horizontal centering is **not required** if a multi-column dashboard layout is used.

- **Border Hierarchy:** A two-level border system must be used:
 - **Dark Lines** (border-gray-500): Main outline and major divisions.
- **Medium Lines** (border-gray-400): Internal grid lines.

4.3. Interactive Embeds

Future reports will rely on Javascript-based interactivity.

- **Containers:** Charts must be injected into named `<div>` containers (e.g., `<div id="chart_main"></div>`).
 - **Hydration:** Scripts should target these IDs for Plotly hydration.
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5. Visualization Standards

5.1. Legacy Static Plots (Matplotlib)

For static image generation (PNG/PDF), Matplotlib remains the engine.

- **Layout Engine:** Use `GridSpec` for any layout more complex than a single subplot. Automatic managers like `tight_layout` are discouraged for complex reports.
- **Implementation Pattern:** Create a "Master Grid" dividing the figure into zones (Title, Plot Area, Legend), then nest sub-grids within those zones.

5.2. Interactive Plots (Plotly) [NEW]

All new interactive visualization features must be built using **Plotly**.

5.2.1. Engine Mandates

- **Graph Objects:** Use `plotly.graph_objects` (`go.Figure`) for maximum control. Avoid `plotly.express` for production reports unless the chart is trivial.
- **Subplots:** Use `plotly.subplots.make_subplots` for multi-panel layouts.

5.2.2. Styling Consistency

- **Style Manager:** All trace styling (colors, fonts, margins) must be retrieved from the project's `PlotlyStyleManager` to ensure visual consistency across the application. Hard-coding hex values in report files is prohibited.

5.2.3. Export Standards

- **Static:** Use `fig.write_image("file.png")` for static snapshots.
- **Web:** Use `fig.to_json()` (or `fig.write_json()`) to export the figure definition for embedding into HTML reports.